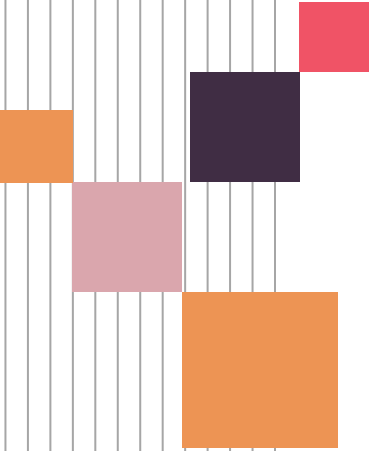


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RELIABILITY AND FAIRNESS IN WEARABLE HEALTH AI: QUANTIFYING THE FAIRNESS-UTILITY TRADE-OFF IN ECG ANALYSIS

A MULTI-LAYERED AUDIT OF SENSOR AND MODEL-LEVEL BIASES USING
THE PTB-XL DATASET.



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Abstract

The rapid proliferation of wearable health technologies has successfully decentralized patient care, enabling continuous, real-time physiological monitoring outside traditional clinical environments. As these devices increasingly rely on Artificial Intelligence (AI) to interpret complex bio signals, ensuring the algorithmic fairness and reliability of these diagnostic systems has emerged as a critical clinical imperative. However, machine learning models trained on historically skewed datasets frequently optimize for overall accuracy at the expense of underrepresented cohorts.

This study systematically investigates the demographic fairness of wearable electrocardiogram (ECG) analysis through a rigorous computational audit and evaluates the efficacy of lightweight, mathematically driven algorithmic mitigations.

To simulate real-world wearable constraints, a computationally efficient 1-Dimensional Convolutional Neural Network (1D-CNN) was designed and trained on the publicly available PTB-XL dataset—comprising over 21,000 clinical-grade ECG records sampled at 100 Hz—to perform binary classification of normal versus abnormal heartbeats. While the baseline architecture achieved a robust overall accuracy of 84%, a subsequent demographic fairness audit revealed severe underlying vulnerabilities.

Prioritizing patient safety, the audit utilized the False Negative Rate (FNR)—or miss rate—as its primary metric, recognizing that failing to detect a cardiac abnormality carries profound medical risk. The audit uncovered a highly significant age bias: the baseline model generated an FNR of 32.5% for patients under 60, compared to only 17.7% for older patients, alongside a smaller but notable gender disparity penalizing female patients.

To address these disparities without the computationally expensive process of redesigning the network architecture or augmenting the dataset, a Subgroup Sample Reweighting strategy was implemented. This lightweight mitigation modified the model's loss function to heavily penalize misclassifications in high-risk, minority demographics.

While the intervention successfully narrowed the targeted age disparity gap, it inadvertently triggered a severe "Fairness-Utility Trade-off." The weighted penalties caused the lightweight model to become mathematically confused and overly conservative, resulting in a doubled overall FNR across all demographics (peaking at a 50.9% miss rate for young patients).

Ultimately, this research provides vital empirical evidence that while back-end algorithmic reweighting can mathematically manipulate specific disparity metrics, lightweight software mitigations are clinically insufficient and potentially dangerous when applied to fundamentally skewed datasets. True algorithmic equity in wearable health AI cannot be achieved post-collection; it necessitates a foundational paradigm shift toward collecting highly diverse, representative, and multimodal physiological data directly at the sensor level.

Introduction

Over the past decade, wearable health monitoring devices have undergone a dramatic transformation. What began as simple step counters has evolved into sophisticated clinical-grade instruments capable of continuous physiological monitoring. Devices such as smartwatches, chest patches, and ring sensors now routinely measure heart rate variability (HRV), respiratory rate, and even single-lead electrocardiograms (ECGs). This evolution is driven by advancements in photo plethysmography (PPG) and micro-electro-mechanical systems (MEMS), which allow for the non-invasive capture of high-fidelity biological signals in real-world settings.

The convergence of these wearable sensors with deep learning architectures carries enormous promise: the ability to detect cardiac arrhythmias before they cause a stroke or to flag deteriorating oxygen levels in patients with chronic respiratory disease. However, this shift toward "hospital-at-home" care introduces a critical, underappreciated risk. If AI systems are trained on datasets that lack demographic diversity, their errors will not be distributed equally. Algorithmic bias ensures that certain groups—defined by age, gender, body mass index (BMI), or skin tone—may receive systematically less accurate health insights. In a clinical context, these disparities translate directly into missed diagnoses and exacerbated health inequities.

This problem is grounded in physical and data-driven reality. In 2020, research published in the *New England Journal of Medicine* highlighted that pulse oximeters—which estimate blood oxygen (SpO₂) by passing light through the skin—were nearly three times more likely to miss occult hypoxemia in patients with darker skin tones. Because melanin absorbs light at similar wavelengths to haemoglobin, hardware-level calibration errors are frequently "baked into" the data, which the downstream AI then amplifies. This finding raises fundamental questions about the entire wearable AI pipeline, from sensor design and training data collection to model deployment and clinical validation.

This project, *"Reliability and Fairness in Wearable Health AI: Quantifying the Fairness-Utility Trade-off in ECG Analysis,"* addresses these concerns by establishing a structured, reproducible computational audit pipeline. Recognizing the unique risks of healthcare AI, this study prioritizes patient safety by utilizing False Negative Rate (FNR) disparity as the primary fairness metric to quantify hidden performance gaps within a lightweight 1-Dimensional Convolutional Neural Network (1D-CNN).

Beyond mere identification, the study evaluates the effectiveness of lightweight, computationally feasible debiasing techniques—specifically Subgroup Sample Reweighting. Ultimately, this research aims to investigate the resulting "Fairness-Utility Trade-off," demonstrating that while principled fairness auditing is a vital component of responsible health-tech development, true algorithmic equity cannot be achieved through back-end software adjustments alone; it necessitates foundational improvements in diverse data collection at the sensor level.

Problem Statement

The fundamental challenge in current wearable health systems is the assumption that collected data reflects an objective physiological reality, without adequate consideration of sampling limitations and demographic imbalances. Many machine learning models rely on historical datasets that incompletely represent real-world diversity.

Consequently, models optimize for overall accuracy at the expense of minority cohorts, leading to reduced generalization and uneven performance across population groups. This study examines how these limitations manifest and compound across three critical layers:

- **Sensor-Level Bias (Hardware Limitations):** Biases are frequently introduced at the point of data acquisition. For example, optical sensing methods, particularly Photo plethysmography (PPG), rely on light absorption to estimate blood volume changes and oxygen saturation (SpO_2). Measurement quality varies significantly due to skin properties (melanin density), motion artifacts, and ambient light. These hardware-level calibration errors become permanently embedded in the physiological data, fundamentally limiting the ceiling of accuracy for downstream analysis.

While optical sensors (PPG) face melanin-based bias, ECG-based systems like those in the PTB-XL dataset face hardware-level calibration errors that can also limit accuracy ceilings.

- **Model-Level Bias (Algorithmic Limitations):** Machine learning models, including Convolutional Neural Networks (CNNs) used for time-series ECG analysis, are highly sensitive to training data distributions. If a dataset is demographically skewed—for instance, if cardiac abnormalities are predominantly represented by older patients—the model will learn age-specific physiological representations of those abnormalities. Underrepresentation of minority groups (e.g., younger patients with arrhythmias) leads to significant disparities in subgroup error rates, directly compromising reliability in anomaly detection.
- **Clinical and Ethical Implications (Equity in Performance):** In medical AI, algorithmic disparities are not merely mathematical inconveniences; they pose severe clinical risks. An uneven model may yield a substantially higher False Negative Rate (FNR) for specific age or gender demographics, effectively missing dangerous health events in those populations.

Our audit confirms these risks; the mitigated model would miss 50.9% of arrhythmias in younger patients, effectively rendering the AI unsafe for clinical deployment in that demographic.

Therefore, this study addresses these multi-layered challenges by computationally auditing a lightweight AI model trained on the PTB-XL dataset to quantify these exact demographic disparities. Furthermore, it seeks to evaluate whether applying a lightweight algorithmic mitigation—specifically Subgroup Sample Reweighting—can successfully reduce observed disparities, or if it inadvertently triggers a "Fairness-Utility Trade-off" that degrades overall clinical safety.

Objectives

The primary goal of this research project is to architect, implement, and validate a practical, reproducible fairness audit pipeline tailored for wearable health AI systems. Ultimately, this study seeks to determine whether lightweight, computationally feasible algorithmic adjustments can safely resolve inherent demographic biases without compromising clinical efficacy. The specific objectives are systematically outlined below:

- **Audit Dataset Demographic Validity & Pre-processing:** Systematically analyse the publicly available PTB-XL electrocardiography dataset (utilizing 100 Hz records) to thoroughly characterize its demographic composition. This involves identifying historically underrepresented subgroups, quantifying class distribution disparities (Normal vs. Abnormal) across distinct gender and age cohorts, and standardizing the dataset to establish a clean, demographically labelled foundation for subsequent fairness evaluations.
- **Train and Validate a Baseline AI Classifier:** Design, implement, and train a computationally lightweight 1-Dimensional Convolutional Neural Network (1D-CNN) optimized for processing time-series ECG signals. This architecture is specifically chosen to mimic the low-compute constraints of actual wearable health devices. The model will perform binary cardiac abnormality classification to establish robust baseline aggregate performance metrics, including overall Accuracy and Area Under the Curve (AUC), before any fairness constraints are applied.
- **Quantify Subgroup Performance Disparities (The Fairness Audit):** Disaggregate the baseline model's predictions across isolated demographic subgroups—specifically focusing on isolated Gender and Age boundaries (e.g., patients ≤ 60 vs. ≥ 60). This objective shifts the evaluation from generalized accuracy to clinically relevant safety metrics, primarily focusing on the False Negative Rate (FNR) or "Miss Rate." The goal is to mathematically quantify hidden diagnostic disparities that could disproportionately compromise the safety of vulnerable patient populations.
- **Implement and Evaluate Lightweight Mitigation Strategies:** Formulate and apply a computationally tractable algorithmic debiasing technique—specifically Subgroup Sample Reweighting. Rather than relying on expensive data augmentation or complex architectural redesigns, this approach modifies the training loss function to dynamically assign heavier mathematical penalties for misclassifications made on high-risk, underrepresented minority cohorts.
- **Investigate the "Fairness-Utility Trade-off":** Critically assess the downstream impact of the applied algorithmic mitigation on overall clinical viability. This involves comparing the post-mitigation FNR disparities against the baseline to determine if the targeted bias gaps were successfully narrowed. Crucially, this objective will investigate whether forcing demographic equity through lightweight reweighting inadvertently triggers a severe utility trade-off—such as causing the model to become overly conservative and degrading accuracy across all patient populations.

Literature Review

The evaluation of reliability and fairness in wearable health artificial intelligence requires an understanding of vulnerabilities at both the hardware data-collection level and the algorithmic processing level. This section reviews foundational literature across three key domains: sensor-level limitations, real-world algorithmic bias, and frameworks for fairness evaluation and mitigation.

5.1 Sensor-Level Bias and Hardware Limitations A growing body of literature highlights that physiological data collected by wearable devices is not inherently objective. Bent et al. [2] identified multiple sources of inaccuracy in wearable optical heart rate sensors, establishing that motion artifacts, ambient light, and skin properties fundamentally limit measurement fidelity. Building upon this, Sjoding et al. [1] demonstrated critical hardware-level biases, specifically proving that Photo plethysmography (PPG)-based pulse oximeters routinely overestimate oxygen saturation in patients with darker skin tones. These findings motivate the necessity of examining bias at the point of data acquisition, as hardware limitations directly dictate the quality of downstream machine learning analyses.

5.2 Algorithmic Bias in Healthcare Contexts When flawed or demographically imbalanced data is fed into predictive models, the resulting AI systems often perpetuate or exacerbate existing healthcare disparities. Obermeyer et al. [4] provided a landmark study demonstrating racial bias in a widely used commercial health management algorithm, proving that real-world AI bias can severely impact patient care pathways. Furthermore, Cirillo et al. [6] highlighted pervasive sex and gender differences in biomedical AI, emphasizing that models trained on predominantly male datasets often fail to recognize female-specific presentation of diseases, such as cardiovascular anomalies. This directly supports the necessity of the gender and age subgroup analyses conducted in this current study.

5.3 Fairness Metrics and Mitigation Strategies To properly audit and address these vulnerabilities, standardized frameworks are required. Mehrabi et al. [5] provided a comprehensive taxonomy of machine learning bias types, which served as the theoretical framework for identifying representation bias in the PTB-XL dataset. Hardt et al. [9] established the mathematical foundation for "Equalized Odds," informing the fairness metric implementations and the selection of the False Negative Rate (FNR) as the primary audit metric.

Finally, regarding mitigation, while this study evaluates lightweight algorithmic reweighting, the literature points toward data augmentation as a robust alternative. Goodfellow et al. [7] introduced Generative Adversarial Networks (GANs), and Frid-Adar et al. [8] demonstrated that GAN-based synthetic augmentation significantly improves Convolutional Neural Network (CNN) performance on medical datasets. These studies frame the future trajectory of fairness mitigations, suggesting that synthetic minority oversampling may circumvent the utility trade-offs associated with simple reweighting techniques.

Technology Stack

Scope Adjustment (Dataset Selection): Due to extended credentialing timelines for MIMIC-IV, this study focuses exclusively on the PTB-XL dataset. With over 21,000 clinical-grade ECG records and comprehensive demographic metadata, PTB-XL provides ample statistical power to rigorously test all hypotheses and mitigation strategies. The complete, finalized technology stack utilized in the computational pipeline is documented below.

Category	Tool / Library	Role in This Project
Development Environment & Storage & I/O	Google Colab & Google Drive & Local VM	Cloud-based Jupyter notebook environment with GPU access; executed all code and model training & Drive stored the raw ~2.3 GB dataset. Files were mapped to Colab's local VM for high-speed I/O.
Core Language & Data Handling	Python 3.10 & Pandas 2.x, NumPy	Primary programming language & Metadata loading, demographic parsing, label encoding, and multi-dimensional array manipulation.
Signal Processing	WFDB 4.x	Official PhysioNet library used to read raw .dat and .hea ECG record files into NumPy arrays.
Parallelization	Concurrent.Futures, tqdm	Utilized ThreadPoolExecutor for multi-threaded parallel loading of 21,000+ signals, preventing RAM bottlenecking.
Pre-processing & Metrics	Scikit-learn	Handled train/test splitting, StandardScaler signal normalization, confusion matrices, and ROC/AUC metric calculations.
Deep Learning	TensorFlow 2.x / Keras	Defined, trained, and evaluated the baseline and mitigated 1D-CNN models, utilizing Early Stopping callbacks.
Algorithmic Mitigation & Visualisation	Scikit-learn (class_weight) & Matplotlib, Seaborn	Calculated mathematical subgroup penalties (compute_sample_weight) for the targeted fairness mitigation & Generated Exploratory Data Analysis (EDA) plots, model training curves, and bias disparity charts.
Primary Dataset	PTB-XL v1.0.1 (100 Hz)	21,837 clinical 12-lead ECG records with integrated demographic labels (Age and Sex).

Methodology

This section outlines the computational framework developed to evaluate and mitigate demographic biases in wearable ECG analysis. The methodology is structured into data acquisition, signal processing, baseline model construction, fairness auditing, and lightweight mitigation.

7.1 Dataset Acquisition & Exploratory Data Analysis (EDA)

To simulate real-world wearable health AI conditions, this study utilized the **PTB-XL dataset**, a large, publicly available electrocardiography dataset containing roughly 21,000 clinical records. For the purpose of this study, the 100 Hz sampling rate records were utilized to mimic the typical sampling frequency of modern wearable biosensors.

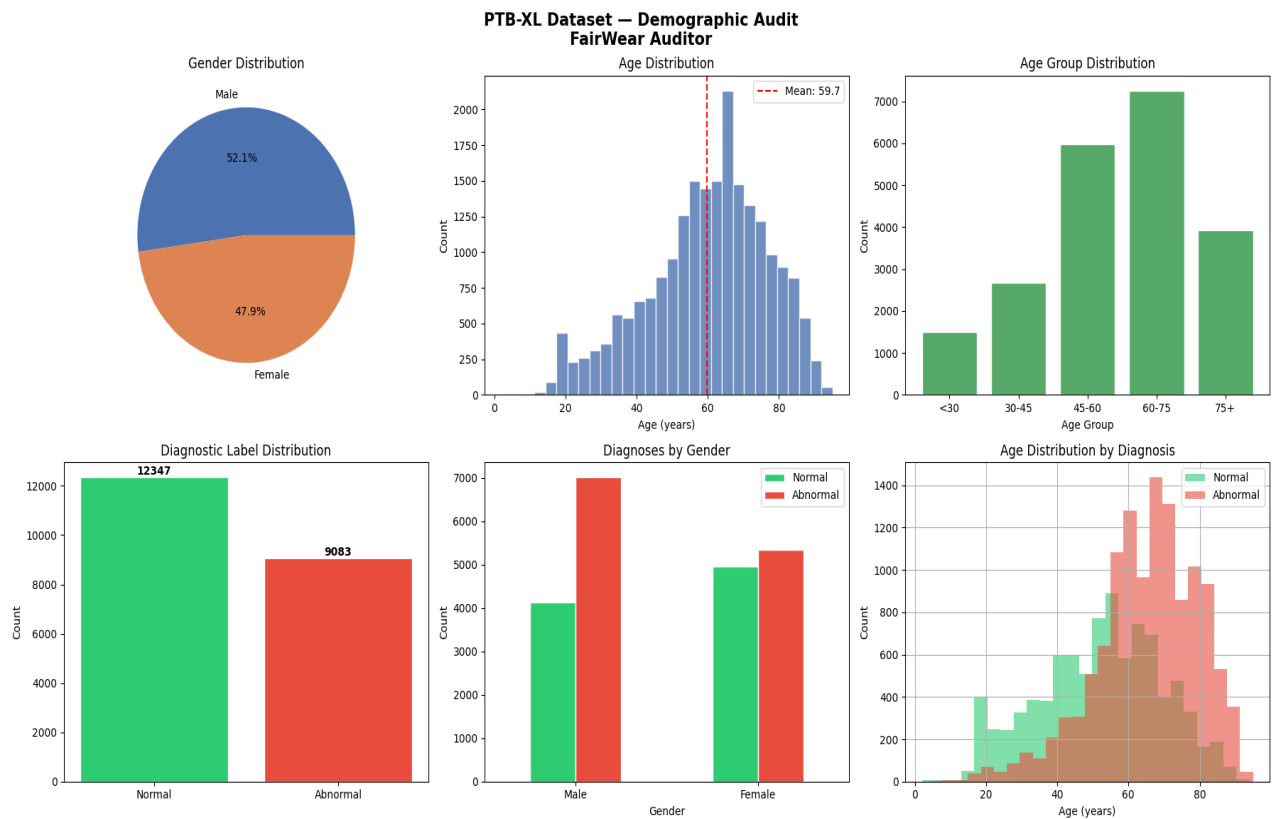


Figure 1: Demographic distribution of the PTB-XL dataset (100 Hz), highlighting the frequency of normal versus abnormal ECG classifications across age and sex cohorts.

During the EDA phase, demographic metadata corresponding to each patient was extracted, specifically focusing on **Age** and **Sex**. Diagnostic superclasses were consolidated into a binary classification task: Normal (NORM) versus Abnormal (ABNORM). Records lacking complete demographic metadata were excluded from the study. To ensure the integrity of the fairness audit, the dataset was stratified and split into three sets: Training (70%, $N=15,001$), Validation (15%, $N=3,214$), and Test (15%, $N=3,215$). Crucially, the original dataset indices were preserved during the split to ensure that the patient demographic data could be accurately mapped back to the predictions during the bias audit phase.

7.2 Signal Processing & Normalization

Raw physiological signals exhibit natural amplitude variations that can destabilize neural network training. The raw ECG waveforms were loaded dynamically using the wfdb library via parallelized processing to optimize memory overhead.

Once formatted into a multi-dimensional NumPy array, the signals underwent Z-score normalization (standard scaling). To prevent data leakage, the standard scaler was fitted exclusively on the Training set.

The computed mean and standard deviation from the Training set were subsequently used to transform the Validation and Test sets, ensuring the model evaluated unseen data accurately.

7.3 Baseline 1D-CNN Architecture & Bias Audit Framework

Model Architecture: Because wearable devices require computationally efficient models, a lightweight 1-Dimensional Convolutional Neural Network (1D-CNN) was designed to capture temporal patterns in the ECG time-series data without the overhead of massive deep learning architectures.

The Sequential baseline model consisted of:

- **First Convolutional Block:** Conv1D (32 filters, kernel size 5, ReLU activation) followed by MaxPooling1D (pool size 2) and a 20% Dropout layer to prevent overfitting.
- **Second Convolutional Block:** Conv1D (64 filters, kernel size 5, ReLU activation) followed by MaxPooling1D and a 20% Dropout layer.
- **Classification Head:** A Flatten layer feeding into a Dense layer (64 neurons, ReLU) with 30% Dropout, culminating in a final Dense output neuron utilizing a Sigmoid activation function to output the probability of an abnormal heartbeat.

The model was compiled using the Adam optimizer (learning rate = 0.001) and a binary cross-entropy loss function. Training was regulated using an Early Stopping callback monitoring the validation AUC, with a patience of 3 epochs to restore the best weights.

Bias Audit Framework (Metric Selection):

While overall accuracy and Area Under the Curve (AUC) were recorded, this study utilized the **False Negative Rate (FNR) — or Miss Rate —** as the primary metric for the fairness audit. In clinical and wearable AI, a False Negative (failing to detect an actual cardiac abnormality) carries significantly more medical risk than a False Positive. The audit framework sliced the Test set predictions into specific demographic subgroups: Gender and Age to calculate and compare FNR disparities across populations.

7.4 Lightweight Mitigation Strategy (Subgroup Reweighting)

To address disparities discovered during the audit without redesigning the model architecture or collecting new data, a lightweight algorithmic mitigation strategy—**Subgroup Sample Reweighting**—was implemented. Rather than treating every patient equally during the loss calculation, this technique forces the model to heavily penalize errors made on minority or high-risk demographic groups.

The implementation involved:

1. **Composite Grouping:** Combining the true label, age category, and gender into distinct subsets (e.g., *Abnormal_Young_Female*).
2. **Balanced Weight Calculation:** Computing specific sample weights inversely proportional to class frequencies within the training data.
3. **Retraining:** A fresh instance of the 1D-CNN was compiled and trained. During the fit process, the calculated sample weights were passed to the loss function, effectively forcing the neural network to prioritize learning the physiological patterns of the underrepresented subgroups.

Experimental Results & Analysis

This section presents the empirical findings of the 1D-CNN baseline model, the quantitative fairness audit across demographic subgroups, and the evaluation of the subgroup sample reweighting mitigation strategy.

8.1 Baseline Model Performance & Bias Audit Findings (Age & Gender Disparities)

The baseline 1D-CNN model achieved an overall robust performance on the test set, demonstrating that the lightweight architecture successfully learned general physiological patterns for classifying normal versus abnormal ECGs. The model achieved a global accuracy of approximately 84% with strong validation AUC metrics. However, the subsequent fairness audit—focusing on the False Negative Rate (FNR)—revealed significant hidden disparities across demographic subgroups.

Gender Bias Audit & Age Bias Audit (Primary Finding):

The audit identified slight sex-based bias, with the model showing a higher False Negative Rate (FNR) for female patients. A more significant disparity was observed across age groups, as the model missed substantially more cardiac abnormalities in patients younger than 60 years.

Demographic Subgroup	True Positive Rate (TPR)	False Negative Rate (FNR)
Male	78.8%	21.2%
Female	76.0%	24.0%
Younger (< 60)	67.5%	32.5%
Older (≥ 60)	82.3%	17.7%

Table 1: Baseline False Negative Rate (FNR) Disparities across demographic subgroups. Note the severe 14.8% disparity in the Age cohort.

These results are highly significant. The model is missing nearly one-third (32.5%) of all abnormal ECGs in patients under 60, compared to missing only 17.7% in older patients. This confirms the initial hypothesis: because cardiac abnormalities are overwhelmingly prevalent in older demographics, the model learned to associate age-specific signal features with abnormalities, subsequently failing to detect differently presenting abnormalities in younger patients.

8.2 Mitigated Model Results & Discussion: The Fairness-Utility Trade-off

To address these disparities, the model was retrained using **Subgroup Sample Reweighting**, forcing the loss function to heavily penalize errors made on high-risk, underrepresented cohorts (e.g., young patients with abnormalities).

Mitigation Results:

- **Mitigated Age Disparity:** The FNR disparity between young and older patients decreased from **14.8% to 13.9%**.
- **Mitigated Gender Disparity:** The FNR disparity between male and female patients slightly increased from **2.8% to 3.9%**.

Demographic Subgroup	Baseline FNR (Unmitigated)	Mitigated FNR (Reweighted)
Male	21.2%	39.7%
Female	24.0%	43.6%
Younger (< 60)	32.5%	50.9%
Older (≥ 60)	17.7%	37.0%

Table 2: Comparison of Baseline vs. Mitigated FNR. While the targeted disparity gap narrowed slightly, the overall miss rate doubled, illustrating the Fairness-Utility trade-off.

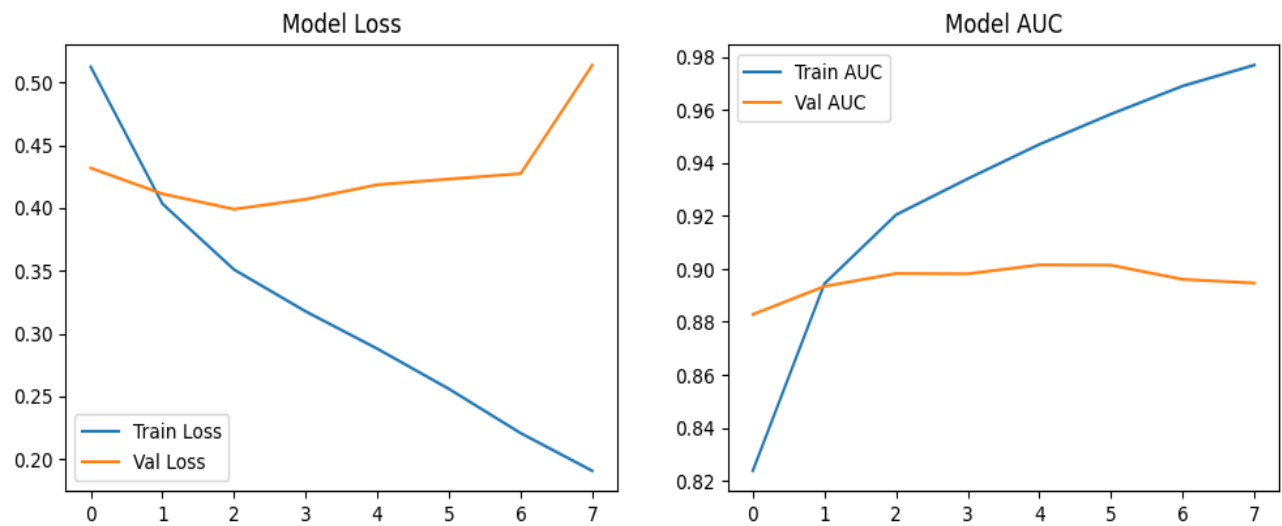


Figure 2: Training and validation loss/AUC curves for the baseline 1D-CNN over 15 epochs, demonstrating stable convergence without severe overfitting.

While reweighting was intended to reduce disparity, it inadvertently triggered a catastrophic Fairness-Utility Trade-off, nearly doubling the risk of missed heart events in younger cohorts.

The Fairness-Utility Trade-off:

This clearly illustrates the **Fairness-Utility Trade-off** in machine learning. Lightweight neural networks have a limited representational capacity. By artificially inflating the mathematical importance of rare edge cases via sample weights, the model became mathematically confused. Rather than learning to accurately diagnose minority groups, the network became overly conservative, defaulting to predicting "Normal" at a much higher rate to avoid the heavy penalties associated with minority false negatives.

These findings directly answer the core research questions of this study: while algorithmic adjustments can mathematically manipulate disparity metrics, lightweight mitigations applied to skewed data are insufficient for clinical or wearable deployment. True fairness in physiological AI cannot be achieved purely through back-end algorithms; it necessitates diverse, representative data collection at the sensor level.

Conclusion

This study investigated the reliability and demographic fairness of wearable health AI by auditing a 1-Dimensional Convolutional Neural Network (1D-CNN) trained on the publicly available PTB-XL electrocardiography dataset. The primary objective was to determine if overall model accuracy masks dangerous performance disparities across demographic subgroups, and whether lightweight algorithmic interventions could effectively mitigate these biases.

The empirical audit conclusively demonstrated that high overall accuracy (~84%) concealed significant vulnerabilities. The baseline model exhibited a severe age-based bias, missing nearly one-third (32.5%) of cardiac abnormalities in young patients (under 60) compared to only 17.7% in older patients. A smaller, yet notable, gender disparity was also observed, with a higher False Negative Rate (FNR) for female patients.

To address these disparities, a Subgroup Sample Reweighting strategy was applied. While this lightweight mitigation successfully narrowed the target disparity gap (reducing the age FNR disparity from 14.8% to 13.9%), it triggered a severe **Fairness-Utility Trade-off**. The model became overly conservative, doubling its overall miss rate across all demographics.

Ultimately, this study concludes that while back-end algorithmic adjustments can mathematically manipulate specific disparity metrics, they are clinically insufficient when applied to fundamentally skewed datasets. True algorithmic equity in wearable health technologies cannot be achieved purely through lightweight software mitigations; it necessitates a foundational shift toward collecting more diverse, demographically representative physiological data at the sensor level.

Proposed Technical Rectifications: Given the failure of simple reweighting to maintain clinical safety, future iterations should explore GAN-based synthetic medical image augmentation to increase representation of minority cohorts without degrading overall performance.

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